Data Science Fundamentals

Chapter 1 — Data, the Data Scientist's Craft & Why Python

Introduction to Data Science

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Today's Roadmap

1. **What is data?**
   structured, semi-structured & unstructured

2. **What is data science?**
   the overlap of three skills

3. **What a data scientist does**
   the lifecycle & the 80/20 rule

4. **Real-world use cases**
   from Netflix to medical imaging

5. **Why Python?**
   the language of data science

+ **Practical, challenges & test**
   code, exercises, self-check

Learning Objectives

By the end of this chapter, you will be able to:

- O1 **Classify** any dataset as structured, semi-structured, or unstructured.
- O2 **Explain** what data science is and **describe** the seven-stage lifecycle.
- O3 **Analyze** the 80/20 rule and justify why data quality drives model quality.
- O4 **Identify** real-world use cases of data science across industries.
- O5 **Justify** the choice of Python and **implement** a first pandas + scikit-learn workflow.

- ***Verbs are measurable (Bloom's): you'll be assessed on classifying, explaining, analyzing, and implementing.***

Why This Matters Now

> *"Learning from data is virtually universally useful. Master it and you will be welcomed anywhere."*
>
> — John Elder, founder of Elder Research

+28%

projected growth in data-science jobs (U.S. BLS, through 2026)

$300–450B

potential annual savings in US healthcare from big-data analytics

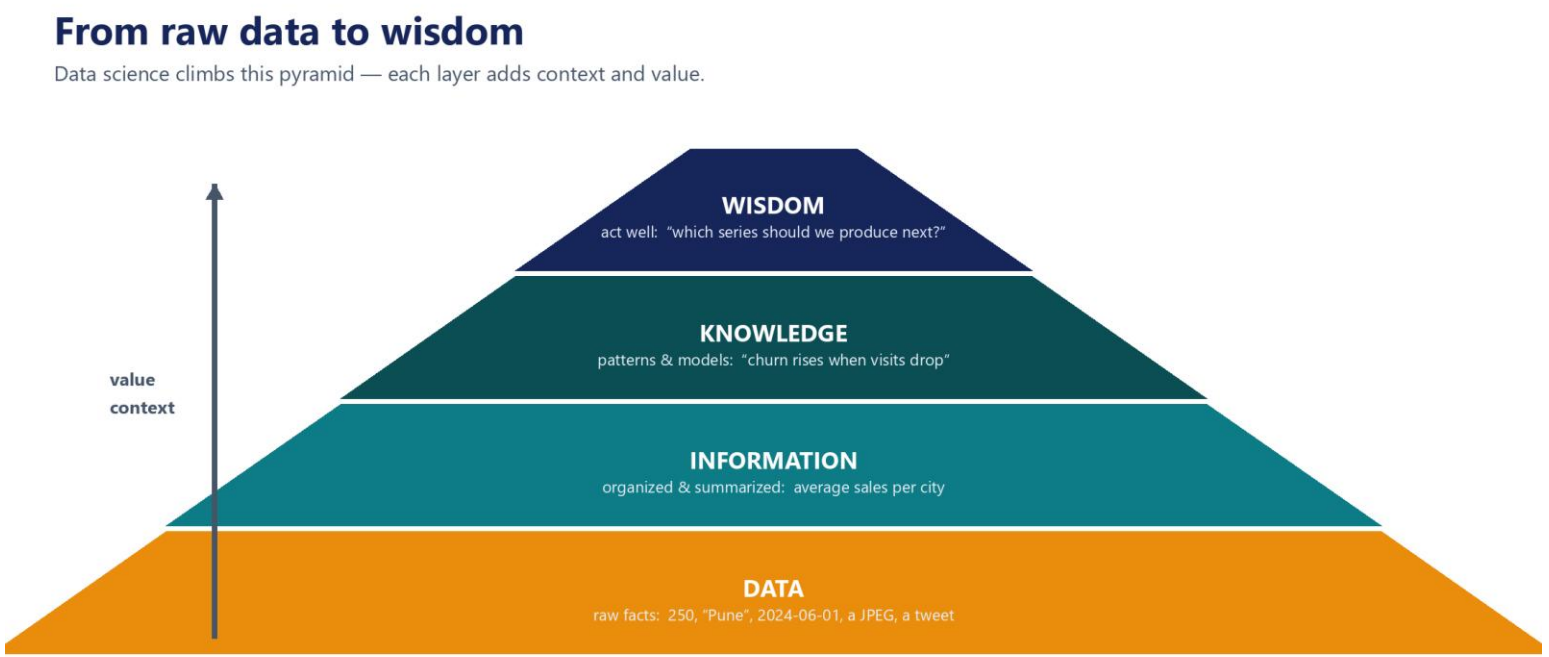
$3.1T

estimated yearly cost of bad / unstructured data to the US economy

What Is Data?

Three categories — and why 80% of it is messy

From Data to Wisdom (DIKW)

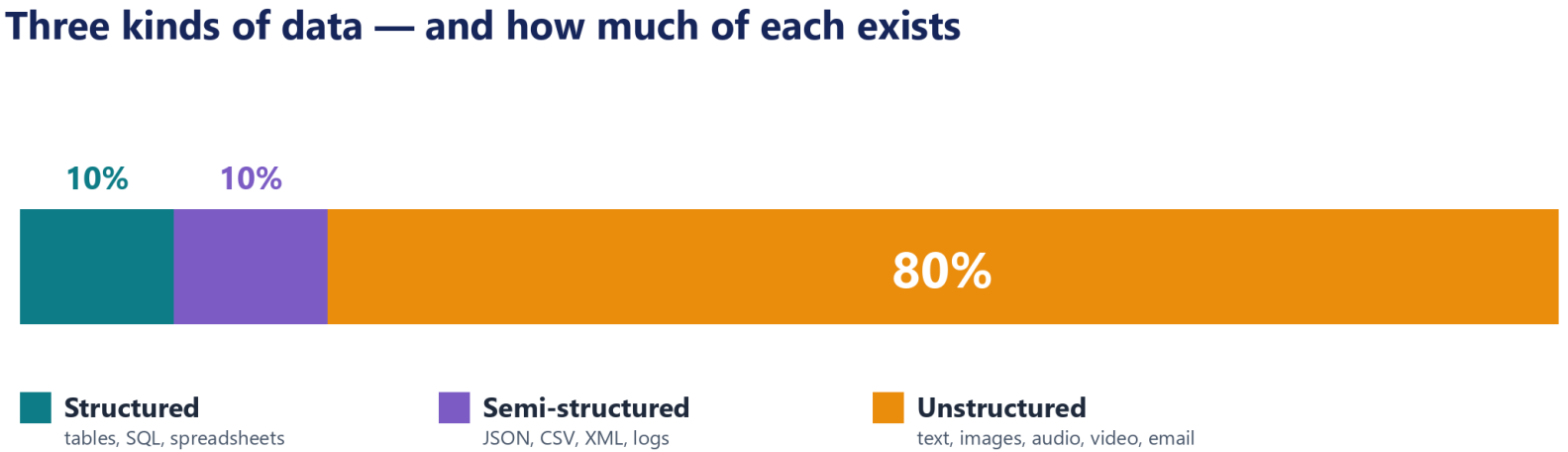


The climb

- **Data** = raw facts, no context
- **Information** = data organized
- **Knowledge** = patterns & models
- **Wisdom** = acting well on it

Data science is the engine that moves you up the pyramid.

Three Kinds of Data



- *Structured ~10% · Semi-structured ~10% · Unstructured ~80%. The hard, valuable work lives in the 80%.*

Structured Data — Neat Rows & Columns

Structured data: a merchants table

merchant_id	name	city	category	revenue
1001	Acme Co	Mumbai	Retail	184,000
1002	Bright Labs	Berlin	SaaS	92,500
1003	Cedar Stores	Pune	Retail	257,300
1004	Delta Foods	Lyon	Food	63,900
1005	Echo Media	Austin	Media	141,200

Each row = one record. Each column = one typed field. Easy to query with SQL / pandas.

Structured

- Well-organized tables, easy to search
- Lives in relational databases (SQL) & spreadsheets
- Every column has one clear type
- Only ~5–10% of the world's data

▸ *If you can put it in a spreadsheet without losing meaning, it's structured.*

Structured Data in Python

PYTHON

structured.py

```python
import pandas as pd

# Structured data: neat rows & columns (like a SQL table)
df = pd.read_csv("merchants.csv")

print(df.head(3))
print(df.shape)      # (rows, cols)  ->  (500, 4)
print(df.dtypes)     # every column has ONE clear type

#    merchant_id     name     city   revenue
# 0         1001    Acme Co   Mumbai   184000
# 1         1002   Bright L   Berlin    92500
# 2         1003    Cedar S     Pune   257300
```

What's happening

- `pd.read_csv` loads a table in one line
- `.shape` = (rows, columns)
- `.dtypes` shows each column's type
- A DataFrame is the workhorse of DS

▸ *pandas turns a CSV or SQL result straight into a table you can compute on.*

Semi-structured Data — JSON, CSV, XML

PYTHON

semi_structured.py

```python
import json
import pandas as pd

# Semi-structured: has tags/keys, but nested and irregular
raw = """
[ {"id": 1, "name": "Acme",   "tags": ["gold","eu"]},
  {"id": 2, "name": "Bright", "tags": ["new"]} ]
"""
records = json.loads(raw)            # -> list of dicts
df = pd.json_normalize(records)      # flatten to a table

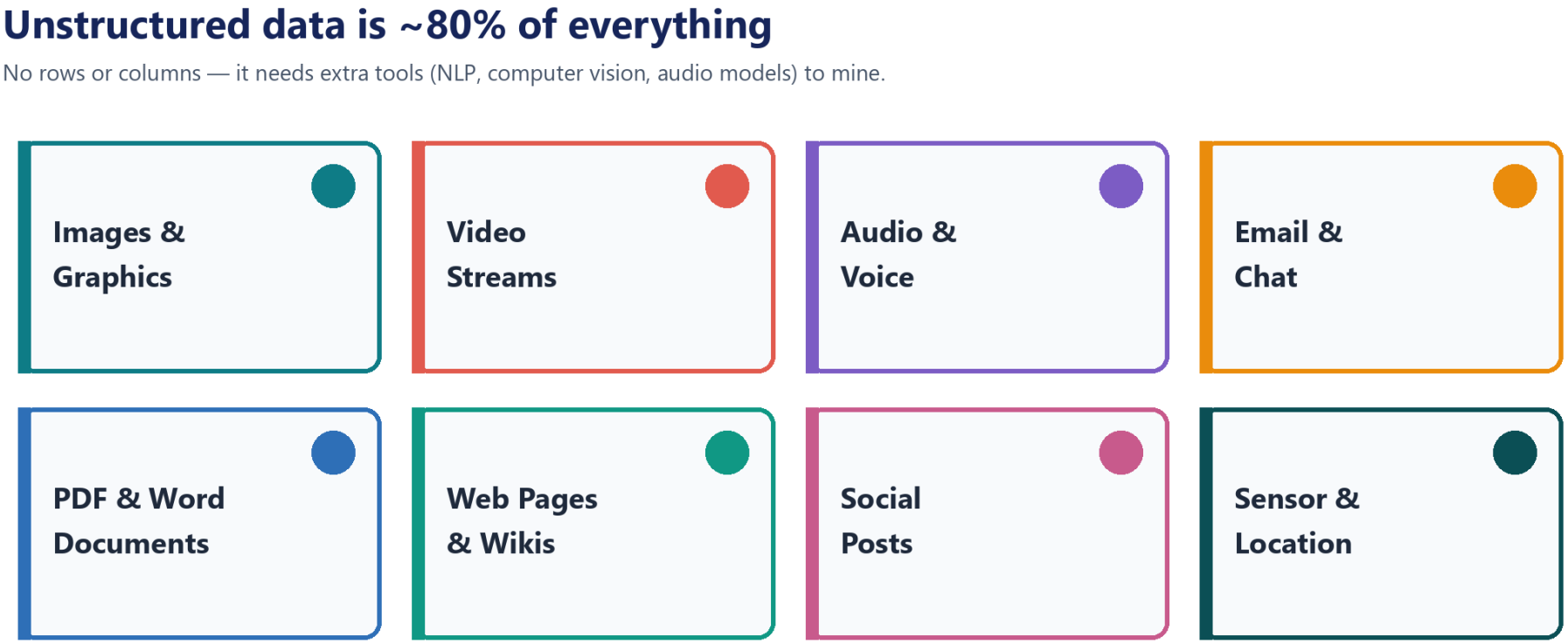
print(df)
#    id    name         tags
# 0   1    Acme    [gold, eu]
# 1   2  Bright         [new]
```

What's happening

- Has tags/keys, but nested & irregular
- **`json.loads`** -> Python dicts/lists
- **`pd.json_normalize`** flattens it
- The 'language of the web & APIs'; ~10%

- *Semi-structured = self-describing but not tabular. Parse it, then normalize into a table.*

Unstructured Data — the Other 80%



- *No rows or columns — images, audio, video, and text need NLP, computer vision & audio models to mine.*

Structured vs Semi vs Unstructured

	Structured	Semi-structured	Unstructured
Examples	SQL tables, spreadsheets	JSON, CSV, XML, logs	images, audio, video, text
Schema	fixed, rigid	flexible / self-describing	none
Share of data	~10%	~10%	~80%
Search / query	very easy (SQL)	moderate (parse first)	hard (needs ML)
Typical tools	SQL, pandas	json, pandas, XML	NLP, CV, deep learning

- *Ease of use falls, and volume + value rise, as you move left → right.*

Practical — All Three in Python

PYTHON

three_types.py

```python
import json
import pandas as pd
from pathlib import Path

# 1) STRUCTURED  -> straight into a table
sales = pd.read_csv("sales.csv")

# 2) SEMI-STRUCTURED  -> parse, then normalize
cfg = json.loads(Path("config.json").read_text())

# 3) UNSTRUCTURED  -> raw bytes, needs extra tools
photo = Path("photo.jpg").read_bytes()
print(len(photo), "bytes of pixels")   # NOT a table

# ~80% of real-world data looks like case 3
```

What's happening

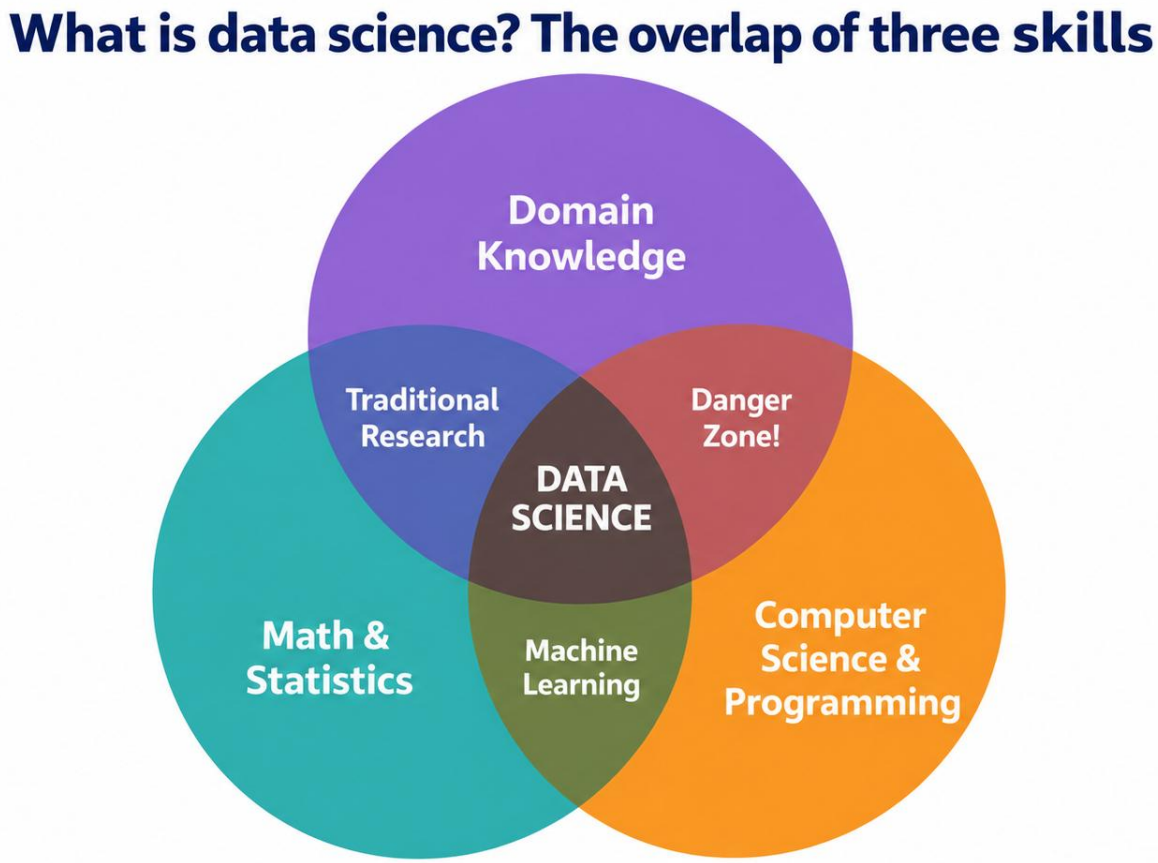
- One script, three data worlds
- `read_csv` → structured table
- `json.loads` → semi-structured
- `read_bytes` → unstructured blob
- Notice: only case 1 is 'ready to analyze'

▸ *Same Python, very different effort: the raw image is just bytes until a model turns it into features.*

What Is Data Science?

Extracting knowledge & insight from raw data

Data Science = an Overlap of Skills



The definition

Data science extracts knowledge & insight from raw data using **math/statistics**, **computer science**, and **domain knowledge**.

- Stats + CS, no domain = 'danger zone'
- Domain + CS, no stats = classic ML
- All three = data science

The Business Value of Data

Modern businesses are awash with data — the winners turn it into decisions.

$300–450B

possible annual US healthcare savings from big-data initiatives (McKinsey)

12–17%

of the $2.6T US healthcare baseline that analytics could reduce

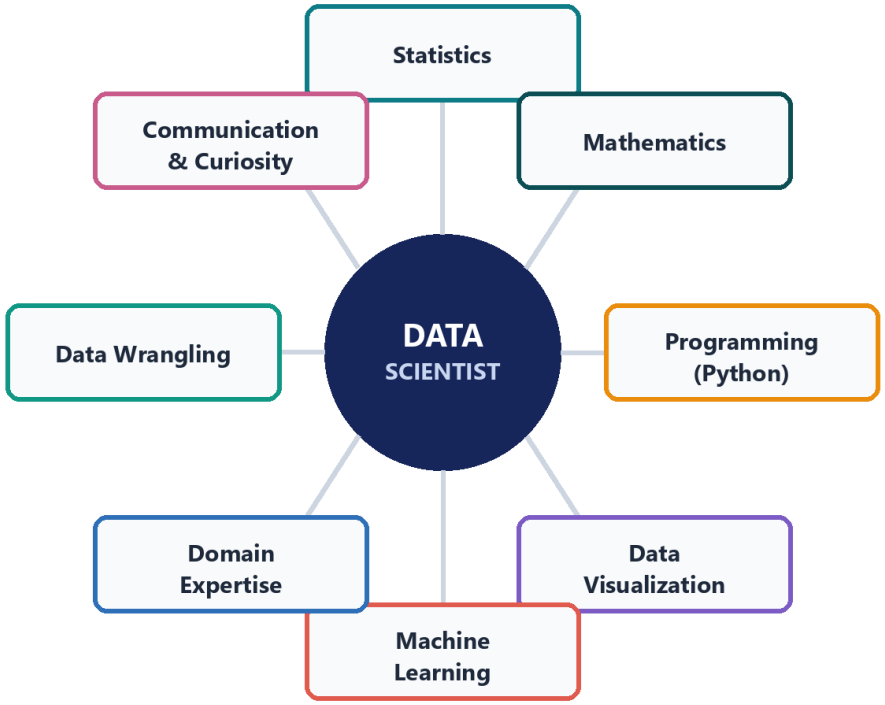
$3.1T / yr

estimated US cost of bad, unorganized data

- ***Data-driven decision-making is now a competitive necessity, not a nice-to-have.***

Skills of a Modern Data Scientist

Skills of a modern data scientist



More than code

- Math & statistics for rigor
- Programming to scale it
- Visualization to communicate
- Domain expertise for relevance
- Curiosity to ask the right question

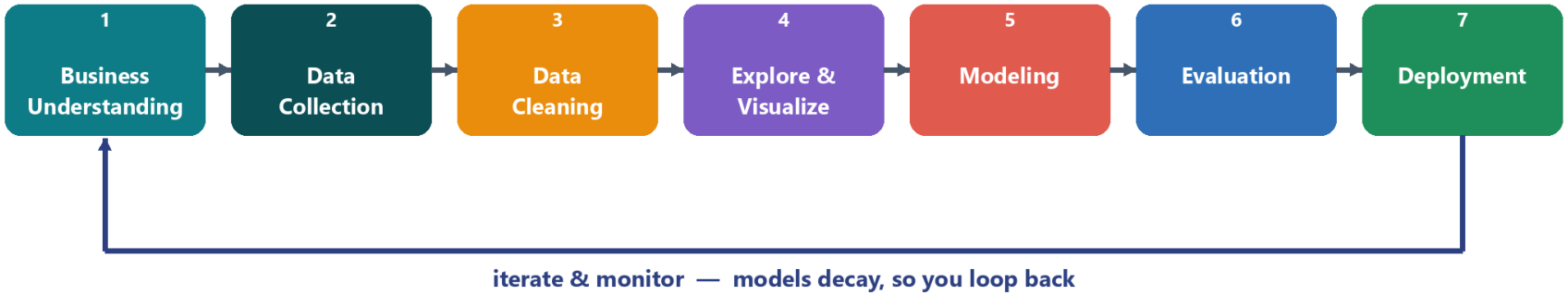
What Does a Data Scientist Do?

A repeatable lifecycle — and a surprising 80/20 split

The Data Science Lifecycle

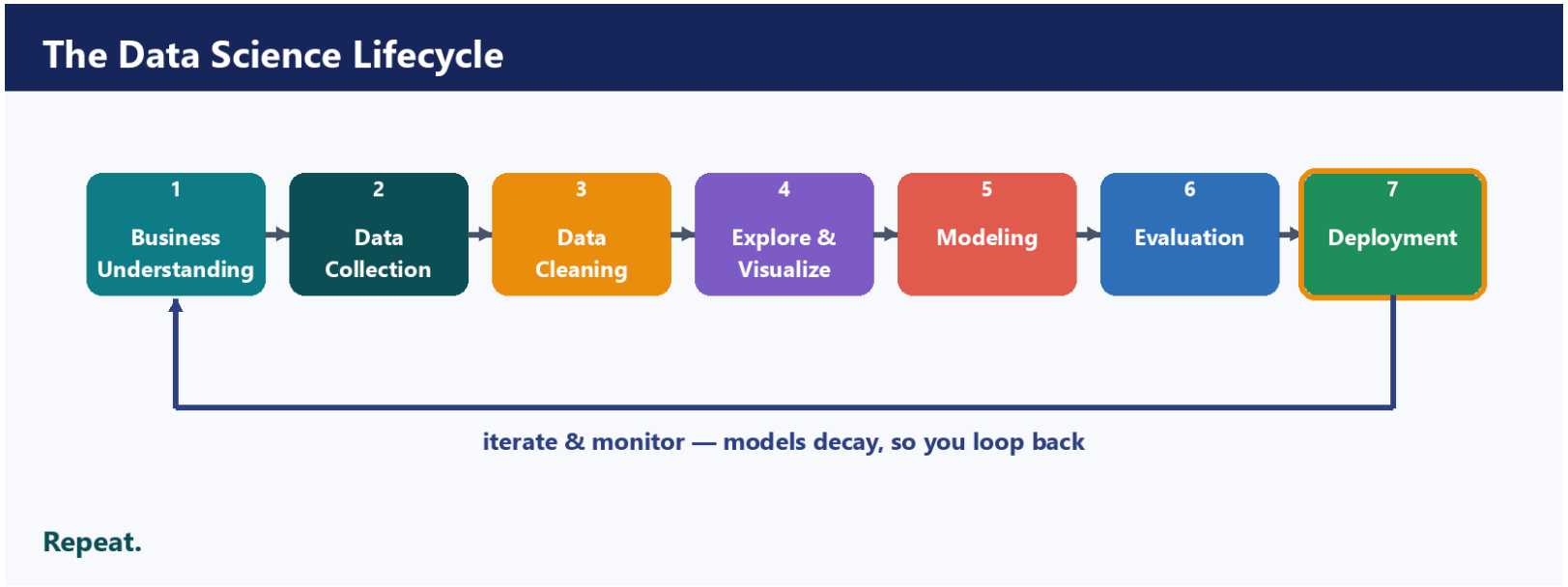
The data science lifecycle

A repeatable cycle — and 80% of the effort is in the first three stages.



- *It starts by asking the right question — and it loops: models decay, so you monitor and iterate.*

See It Move — the Lifecycle in Motion



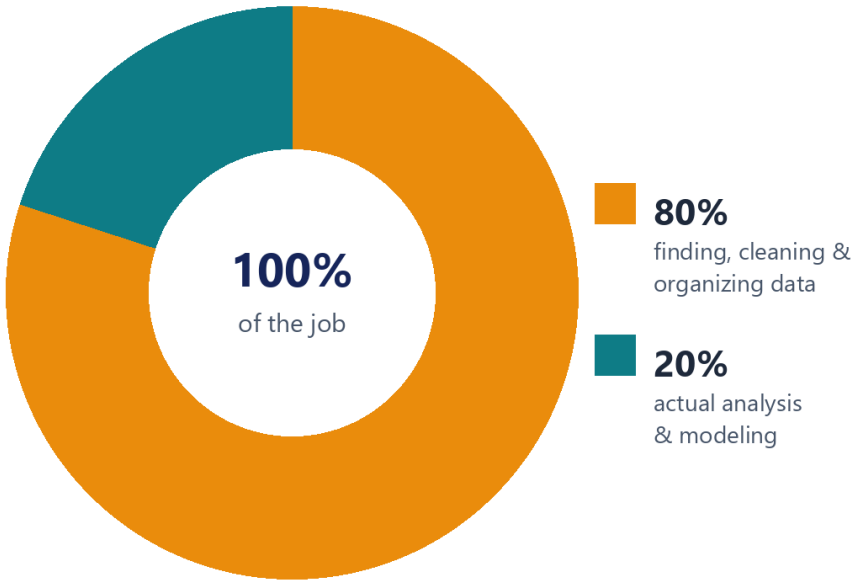
▶ Click to play

Watch for

- stages activate in order
- the first 3 are the heavy lifting
- the arrow loops back
- no 'done' — only iterate

The 80/20 Rule of Data Science

Where the time really goes



"A model is only as good as the data used to build it."

Garbage in, garbage out

- ~80% of time: finding, cleaning, organizing data
- ~20%: the actual analysis & modeling
- Tedious — but decisive
- A model is only as good as its data

Practical — Your First DS Project

PYTHON

first_ds_project.py

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# 1. LOAD          2. CLEAN
df = pd.read_csv("customers.csv").dropna()
X = df[["age", "income", "visits"]]
y = df["churned"]

# 3. SPLIT
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2)

# 4. MODEL         5. EVALUATE
model = LogisticRegression().fit(X_tr, y_tr)
pred  = model.predict(X_te)
print("accuracy:", accuracy_score(y_te, pred))   # ~0.86
```

What's happening

- The whole lifecycle in ~12 lines
- **load → clean → split**
- **fit → predict → score**
- `scikit-learn` = models in 1 line
- Change the CSV, keep the recipe

- *This template — load, clean, split, model, evaluate — solves a huge share of real problems.*

Real-World Use Cases

“Information is the oil of the 21st century.”

Data Science Is Everywhere

Google
Object detection in images

Amazon
Product recommendation engine

Netflix
Viewing patterns → which originals to produce

Baidu / RSNA
Detect tumors & pneumonia in medical scans

Airbus
Spot ships in satellite imagery

Santander
Predict transaction value per customer

YouTube
Automated video classification

IDB Bank
Predict household need for aid targeting

- ***Shipping, healthcare, finance, media, aviation — no domain is untouched.***

Deep Dive — Recommendation Systems

Amazon — “customers also bought”

- Idea: people with similar histories like similar things (collaborative filtering)
- Input: a giant, sparse user × item matrix
- Output: top-N items you haven't seen yet
- Impact: a large share of sales comes from recommendations

Netflix — what to produce next

- Mines viewing patterns of millions of users
- Predicts what themes/actors will engage
- Informs which original series to greenlight
- Data doesn't just recommend — it creates content

Why Python for Data Science?

The Swiss-army knife of the coding world

Why Python Wins for Data Science

Beginner-friendly

Reads almost like English — low barrier to entry

Multi-paradigm

OOP, functional & procedural — flexible for any task

Free & open-source

Anyone can extend it; no license cost

Batteries + libraries

pandas, NumPy, scikit-learn already tailored to DS

Huge community

Answers, tutorials & packages everywhere

“2nd-best at everything”

Glue language that connects the whole stack

- ***Simplicity + a rich analytical ecosystem = the default choice for data work.***

The Python Data Science Stack

NumPy
fast arrays & math — the foundation

pandas
tables (DataFrames) & wrangling

Matplotlib / Seaborn
charts & visualization

scikit-learn
classic ML models & pipelines

Jupyter
interactive notebooks for exploration

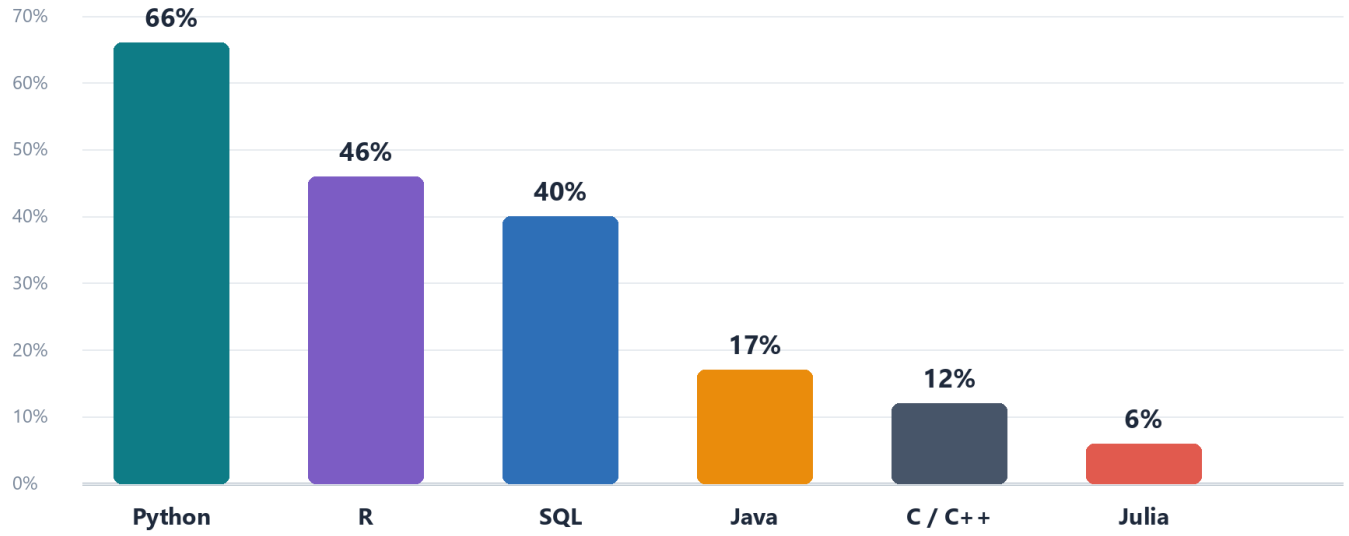
TensorFlow / PyTorch
deep learning for the 80%

- *You'll meet these across the course — pandas first, then scikit-learn, then deep learning.*

Python Leads the Field

Preferred language for data science / ML

Python leads by a wide margin (illustrative, KDnuggets-style poll).



The survey says

- Python is the clear #1 for DS/ML
- R stays strong for statistics
- SQL is essential for data access
- Learn Python first, add SQL early

Practical — A pandas Quick Tour

PYTHON

pandas_tour.py

```python
import pandas as pd

s  = pd.Series([10, 20, 30], name="price")
df = pd.DataFrame({
    "city":  ["Pune", "Berlin", "Pune", "Berlin"],
    "sales": [250, 300, 175, 420],
})

df["sales"].mean()                  # 286.25
df[df.sales > 200]                  # filter rows
df.groupby("city")["sales"].sum()   # Pune 425, Berlin 720

# One library covers most of your data wrangling.
```

What's happening

- `Series` = one column
- `DataFrame` = a whole table
- Filter rows with a boolean mask
- `groupby` = split-apply-combine
- 90% of wrangling is these four moves

- *pandas is the single most important library in this course — get comfortable here.*

Your Turn — Set Up the Toolkit

Before next session

- Install **Anaconda** (Python + libraries in one)
- Launch **Jupyter Notebook**
- Run `import pandas as pd`
- Download a small CSV to explore

What's coming in Chapter 2

- System setup & installing Anaconda
- Knowing your Jupyter notebook
- Installing new Python libraries
- Then: lists, dictionaries & real data

▸ *LEARN → APPLY → REPEAT. The lab manual walks you through all of this, step by step.*

Coding Challenges

Four problems — from warm-up to project sketch

Challenge 1 — Classify the Data

CHALLENGE 1

EASY

Label each example by its data type

For each item, answer **structured**, **semi-structured**, or **unstructured**:

a) `a MySQL table of orders`

b) `a folder of .mp3 files`

c) `an API response in JSON`

d) `scanned PDF invoices`

e) `a CSV export`

f) `customer support chat logs`

Then: *estimate what % of a typical company's data is each type.*

- ***Answers: a-structured, b-unstructured, c-semi, d-unstructured, e-semi/structured, f-unstructured.***

Challenge 2 — Explore a Dataset

CHALLENGE 2

MEDIUM

Profile a CSV with pandas

Load any CSV and write **`explore(path)`** that returns a dict with:

- `rows` and `cols` (from `.shape`)
- `missing` = total count of NaN cells (`.isna().sum().sum()`)
- the dtype of each column (`.dtypes`)

Stretch: *print the 3 columns with the most missing values.*

- ***This 'profile first' habit is the real first step of every project.***

Challenges 3 & 4

CHALLENGE 3

MEDIUM

JSON → DataFrame

Read a nested `.json` file and flatten it into a tidy table.

Use `json.load` + `pd.json_normalize`.

Handle a list of records with nested fields (e.g. `address.city`).

CHALLENGE 4

HARD

Sketch a DS Workflow

Pick a business problem (e.g. “predict customer churn”).

Write the 7 lifecycle stages for it, naming for each: the data you'd collect, and how you'd measure success.

Challenge Starter Code

ch1_challenges.py

```python
# --- Chapter 1 challenges: fill in each body ---

def classify(example: str) -> str:
    """C1: 'structured' | 'semi' | 'unstructured'."""
    ...

def explore(csv_path: str) -> dict:
    """C2: return {rows, cols, missing} for a CSV."""
    ...

def json_to_frame(json_path: str):
    """C3: load JSON -> a flat pandas DataFrame."""
    ...

def sketch_workflow(problem: str) -> list:
    """C4: return the 7 data-science stages in order."""
    ...
```

How to work

- Fill in one function at a time
- Test with a tiny CSV / JSON you make
- Write the base/simple case first
- Full signatures in the speaker notes

Test Yourself

Part A: multiple choice · Part B: short answer

Test — Part A: Multiple Choice

1. Which is an example of structured data?

a) a JPEG photo b) a SQL table c) an audio clip d) a PDF

2. About what share of data is unstructured?

a) 10% b) 25% c) 50% d) 80%

3. JSON, CSV and XML are examples of:

a) structured b) semi-structured c) unstructured d) binary

4. Data science mainly extracts ___ from data.

a) electricity b) knowledge & insight c) hardware d) syntax

5. ~80% of a data scientist's time goes to:

a) modeling b) deployment c) cleaning data d) meetings

6. Which language leads the KDnuggets DS poll?

a) Java b) R c) Python d) C++

Test — Part B: Short Answer

B1. Define data science in one sentence, naming the three skill areas it combines.

B2. Give one real example each of structured, semi-structured, and unstructured data.

B3. List the seven stages of the data science lifecycle, in order.

B4. Explain the 80/20 rule and why data quality limits model quality.

B5. Give two reasons Python is the popular choice for data science.

Test — Answer Key & Rubric

Part A — Multiple Choice

✓ 1 — b (SQL table)

✓ 2 — d (80%)

✓ 3 — b (semi-structured)

✓ 4 — b (knowledge & insight)

✓ 5 — c (cleaning data)

✓ 6 — c (Python)

Part B — Model Answers

B1: Extracting knowledge/insight from raw data using math/statistics, computer science, and domain knowledge.

B2: e.g. SQL table / a JSON API response / a photo or audio clip.

B3: Business understanding, data collection, cleaning, explore & visualize, modeling, evaluation, deployment.

B4: 80% of effort is data prep; a model can't exceed the quality of its data — garbage in, garbage out.

B5: Simple readable syntax + a rich free library ecosystem (pandas, scikit-learn); large community.

Key Takeaways

- ✓ Data comes in three kinds **— and ~80% is unstructured.**
- ✓ Data science turns raw data into knowledge, insight & decisions.
- ✓ The work is a lifecycle — and 80% of it is cleaning data.
- ✓ It's everywhere — healthcare, finance, media, aviation, retail.
- ✓ Python is the default tool — pandas is where you start.

The only rule you need: LEARN → APPLY → REPEAT.

References & What's Next

Go deeper

- *pandas documentation*
- *scikit-learn: getting started*
- *KDnuggets — data science news*
- Course textbook — Chapter 1: Data Science Fundamentals

Next: Chapter 2 — Setup

- Install & configure Anaconda
- Master the Jupyter notebook
- Add libraries with conda / pip
- Then: Lists & Dictionaries (Ch. 3)

Do the lab manual (Sessions 1 & 2) before the next lecture.